

Supplementary Information

Few biological parameters suffice to reproduce measured dynamics in systems biology models:
local sensitivity-based reduction with formally verified mathematical guarantees

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Supplementary Table S1. Selected biological parameters

Final subset S for each problem (after backward elimination) and the parameters removed by backward elimination. Calibration parameters, which are always re-estimated, are not listed. Crauste and Beer have no admissible subset; the subset at which the method stopped is shown. For Crauste it is not meaningful: about half of the perturbed simulations have no solution on the measured horizon (finite-time growth without bound), so the sensitivity matrix is incomplete.

System	Selected biological parameters S	Removed by backward elimination
Alkan	kt, k_rep_nhej	–
Armistead	k_d, alpha_cer	k00, k2
Bachmann	CISRNADelay, JAK2ActEpo	–
Beer	kdeg_i_typeIDT1, kdeg_i_typeIDT3, ksyn_typeIDT3, kdim_typeIDT1, beta_typeIDT3_ExpID2, beta_typeIDT3_ExpID3, beta_typeIDT3_ExpID5, beta_typeIDT3_ExpID1, beta_typeIDT3_ExpID4, init_Bac, beta_typeIDT1_ExpID6, beta_typeIDT1_ExpID5, beta_typeIDT1_ExpID1, beta_typeIDT1_ExpID4, beta_typeIDT1_ExpID3, kdim_typeIDwt, ksyn_typeIDT5, kdim_typeIDT5	–
Bertozzi	R0_NY, I0_NY	–
Blasi	a_basal, a_k16_k12k16	a_k12k16_k8k12k16, a_k5k12_k5k8k12, a_k8k12k16_4ac
Boehm	k_imp_hetero, k_phos	–
Borghans	Kz, K_par	–
Brannmark	k1f, km2	–
Bruno	kb1, init_bcar2, init_zea_1, init_bcar1, init_bcry_1	szea, init_b10_1
Chen	k106, k44	–
Crauste	rho_E	–
Elowitz	n_Hill, tau_mRNA	tau_prot, tau_prot_GFP, eff
Fiedler	tau2, k6	–
Fujita	init_AKT, reaction_6_k1	–
Giordano	alpha_4, alpha_28	–
Isensee	kf_RIIp_C_2_RII_C_2, xi_kf_RII_C_2_RII_2	ks_AC_cAMP, KD_cAMP, xi_AC_cAMP_Fsk, kf_RIIp_cAMP_C_2_RIIp_2, kf_RII_2_RII_C_2
Lang	kDeE2f1, kDeCe	–
Laske	k_imp, k_syn_R_M	k_bind_M1
Liu	kappa, phi	–

System	Selected biological parameters S	Removed by backward elimination
Lucarelli	init_Rec, S2tot, S3tot, S_dephosphos, S_phos, geneF_turn, k_234, geneK_turn, S4tot, geneH_turn	–
Okuonghae	transmission_rate_effective, gamma_i	–
Oliveira	beta_0, h_hosp_rate	–
Perelson	c, delta	–
Rahman	infected_moderate_transmission_rate, infected_normal_worsen_rate	–
Raia	SOCS3mRNA_production, init_Rec_i	–
Raimundez	kdeg_membran_MKN1, d_kimp_pEGFR_EGF_2_MKN1_2_HS746T	ka_MAPK_MKN1
SalazarCavazos	ratio_kpkd_Y1068_FREE, ratio_kpkd_YN_FREE	–
Schwen	ini_R1, ini_R2fold	–
Smith	k8, kminus7b	–
Sneyd	l4, l2	–
Weber	s12, a21	–
Zhao	R_Stage_I_China, R_Stage_II_Wuhan, gamma_1_Stage_I_Hubei	–
Zheng	k01_02, k13_12	–

Supplementary Table S2. Notation

Symbol	Meaning	Value used
$x(t; \theta) \in \mathbb{R}^n$	state trajectory of the model $\dot{x} = f(x, \theta)$	–
$\theta \in \mathbb{R}^p, \theta_0$	parameter vector; nominal (published) value	PEtab nominal values
Θ_B, Θ_C	biological parameters (in the SBML model) and calibration parameters (observation scalings and offsets)	from the PEtAb tables
$y_i(\theta) = h(g_i(x(t_i), \theta))/\sigma_i$	i -th scaled output: observable g_i , transformation h (lin, log, log ₁₀), noise s.d. σ_i , at the condition and time of measurement i	PEtab measurement table
$J = \partial y / \partial \log \theta \in \mathbb{R}^{m \times p}$	relative sensitivity matrix (central differences, step δ)	$\delta = 0.01$
$E_j = \ J_{:,j}\ ^2$	sensitivity energy of parameter j	–
$R_{\text{var}}(S)$	fraction of the energy of biological parameters captured by S	$\tau_R = 0.89$
s_j	finite-perturbation effect of parameter j (stage-2 ordering)	$\eta = 0.1$
$\kappa(S), \text{VIF}(S)$	condition number and largest variance inflation factor of the normalised columns of J_S	$\tau_\kappa = \tau_{\text{VIF}} = 10$
$\Delta y_{\text{full}}, \Delta y_S$	output change when all parameters, or only those in $S \cup \Theta_C$, take their scenario values	$N = 15, \varepsilon = 0.05$
$\cos \Delta$	cosine similarity between Δy_{full} and Δy_S	–
e_{rel}	relative error without re-estimation	–

Symbol	Meaning	Value used
e_{fit}	relative error after re-estimating θ_S and θ_C	–
$\tau_e = \sqrt{1 - c^2}$	admissibility threshold on the median e_{fit}	$c = 0.9$, $\tau_e = 0.436$
k_{min}	minimum subset size	2
χ^2/n	mean squared noise-scaled residual on test data	–
e_{pred}	relative prediction error with respect to a known truth (synthetic experiment)	–

Supplementary Table S3. Formal theorems

Correspondence between the statements of Table 3 of the main text and the Lean 4 theorems (repository folder `lean/`). All files compile with Lean 4.23.0 and Mathlib without `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

Statement	Lean theorem(s)
Energy and R_{var}	<code>FirstOrderErrorBound.total_eq_kept_add_removed</code> , <code>first_order_error_bound</code> , <code>first_order_error_from_Rvar</code>
Local trajectory bound	<code>CertifiedODEReduction.certified_parameter_reduction</code>
Sensitivities of the ODE	<code>ODEParamDiff.hasFDerivAt_solution_param_init</code>
Greedy selection	<code>GreedySelection.greedy_kappa_vif_guarantee</code>
Identifiability	<code>Identifiability.ls_unique</code> , <code>ls_param_kappa_stability</code> , <code>ls_vif_inverse_stability</code>
Theorem 1 (admissibility threshold)	<code>ScaledError.min_rel_error_eq</code>
Re-estimation	<code>RefitMonotone.fitErr_anti</code> , <code>fitErr_union_le</code> , <code>fitErr_le_ere1</code> , <code>admissible_mono</code>
Theorem 2 (backward elimination)	<code>Poda.prune_spec</code> , <code>Poda.exists_minimal_admissible</code>
Measured outputs	<code>OutputComposition.output_sq_bound</code> , <code>output_uniform_bound</code> , <code>log_lipschitz</code> , <code>log10_lipschitz</code>
First-order e_{fit}	<code>LinearizedFit.lin_le_scaled</code> , <code>fit_upper_of_lin</code> , <code>fit_lower_of_lin</code>
Finite-difference and numerical margins	<code>NumericalRobustness.fd_column_error</code> , <code>rvar_certified_from_approx</code> , <code>kappa_certified_from_approx</code> , <code>vif_certified_from_approx</code>
Model assumptions (22 models)	<code>KineticNetwork.network_final</code> , <code>StrictNetwork.strict_final_init</code> , <code>RiccatiNetwork.riccati_final</code> ; model files in <code>lean/Models/</code>

Supplementary Note. Verification of the simulator

Our simulation of the measured outputs was compared, at the nominal parameters, with the reference simulations distributed with the benchmark, matching each measurement by observable, condition and time. For the problems with comparable reference files the median relative difference was below 10^{-5} in all but five cases: Chen (1.1×10^{-4}), Zheng (whose reference file contains the measured data rather than a simulation), Crauste (0.8%, very stiff model), Lang (2.8%, an oscillating cell-cycle model simulated to $t = 136,472$, where peaks differ by an accumulated phase shift) and Perelson (17%; the initial state of the model is stationary and our simulation respects it, which suggests that the reference was generated with different values). Several problems have no comparable reference (no file, or different condition identifiers). The full comparison is given in the repository (`certificados/resultados/salidas/VERIFICACION_SIMULADOR.md`).